Text Mining Analysis of Glassdoor Job Reviews

University of Connecticut School of Business

Course OPIM 5671

Data Mining and Time Series Forecasting

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*The work contained and presented here is the team 's work and the team 's work alone.

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Executive Summary:

This project presents a comprehensive text mining and sentiment analysis of employee reviews collected from Glassdoor. Focusing on both structured and unstructured data across various companies, industries, and job roles, the study aims to uncover the key factors driving employee satisfaction and dissatisfaction. Using natural language processing (NLP) techniques, including sentiment classification, topic modelling, and predictive modelling, we analyze review components such as "pros," "cons," and overall ratings to extract meaningful themes.

Reviews are classified into positive, neutral, or negative sentiment categories, and predictive models are built to forecast employee satisfaction based solely on textual content. The best-performing models are selected based on accuracy and interpretability, ensuring a balanced trade-off between performance and insight. The study further investigates correlations between review sentiment and structured attributes like work-life balance, compensation, and CEO approval.

Insights derived from this analysis are valuable for human resource professionals aiming to improve workplace culture, for companies benchmarking against industry peers, and for job seekers seeking transparent, experience-based information. The findings demonstrate the effectiveness of text mining in capturing and forecasting workplace sentiment, with potential applications in HR strategy and career decision-making.

Introduction:

Employee satisfaction has emerged as a critical factor influencing organizational success, workforce stability, and overall productivity. In the digital age, platforms like Glassdoor have provided a transparent space for employees to share their workplace experiences, offering valuable insights into company culture, leadership effectiveness, and job satisfaction. This project seeks to analyze large-scale employee review data from Glassdoor, focusing on both structured ratings and unstructured textual feedback, to uncover trends and patterns in employee sentiment across companies, industries, and job roles.

Using natural language processing (NLP) and text mining techniques, the study aims to extract key themes from reviews, classify sentiment, and explore correlations between written feedback and structured metrics such as work-life balance, compensation, and management satisfaction. The project will also evaluate how factors like CEO approval and recommendation status influence overall employee ratings. Furthermore, we aim to build predictive models to determine whether review text alone can accurately anticipate an employee's overall satisfaction score. By translating qualitative feedback into actionable insights, this research will support organizations in enhancing HR strategies and help job seekers make more informed career choices.

Key Variables in the Dataset:

The dataset includes a mix of structured and unstructured variables that provide valuable insights into employee experiences across different companies, roles, and industries. The following are four key variables used in this analysis:

1. Pros (Textual Review)
Description: This field captures the positive aspects of an employee's experience in their own words. Employees describe what they appreciated most about their job, such as a supportive work environment, career growth, or flexibility.

Significance: The "pros" section is a rich source for identifying positive sentiment and repeated themes employees associate with satisfaction. Text mining this field helps uncover strengths across different companies and roles, informing both organizational development and job seeker expectations.

2. Cons (Textual Review)
Description: This field includes employee-reported challenges or negative aspects of their job, such as poor management, lack of growth, or high workload. It reflects areas where employees were dissatisfied or faced obstacles.

Significance: Analyzing the "cons" section reveals recurring pain points across industries and roles. It provides valuable input for sentiment classification and helps organizations understand which issues most negatively impact employee experience.

3. Overall Rating (1–5)
Description: A structured, numerical rating that reflects the employee's overall job satisfaction. The scale ranges from 1 (very dissatisfied) to 5 (very satisfied).

Significance: This is the target variable in our predictive modeling efforts. It allows us to correlate textual reviews with overall sentiment, test model accuracy, and assess how well written feedback predicts satisfaction.

4. Work-Life Balance (1–5)
Description: This numeric rating evaluates the employee's perception of their ability to balance job responsibilities with personal life. Scores range from 1 (poor balance) to 5 (excellent balance).

Significance: Work-life balance is a frequently discussed theme in reviews and often a key driver of job satisfaction. This variable serves as an important structured indicator that can be correlated with text-based themes extracted from review content.

Data Source: Glassdoor Job Reviews Datasets

Data Cleaning and Preprocessing:

[***LINK TO PYTHON NOTEBOOK***]
To prepare the data for analysis, several preprocessing steps were taken:

1. **Dataset Import and Initial Inspection**
   - Imported Glassdoor review dataset using pandas.read_csv().
   - Displayed column info using .info() and calculated missing value percentages using .isnull().sum().
   - Identified top firms by frequency using value_counts().

2. **Company-Specific Subsetting**
   - Filtered the dataset for selected firms for focused analysis: The four companies — IBM, J.P. Morgan, Oracle, and McDonald's — were selected to ensure diversity across industries (tech, finance, software, and food services) while maintaining a balanced target variable (overall_rating) and a manageable dataset size for modeling efficiency. This mix supports meaningful comparison across domains and enables both supervised and unsupervised text analysis without computational overload.
   - Created separate dataframes: df3 (IBM 60,436 rows), df5 (McDonald's 49,450 rows), df10 (Oracle 31,941), df12 (J.P. Morgan 25,814)

3. **Feature Selection**
   - Retained only relevant columns for analysis using list L1, which includes structured: firm, overall_rating, work_life_balance, culture_values, career_opp, comp_benefits, senior_mgmt, Unstructured: pros, cons, headline, Sentiment targets: recommend, ceo_approv, outlook, Metadata: current, job_title, date_review

4. **Handling Missing Values**
   - Focused on rating columns with potential missing values: ['work_life_balance', 'culture_values', 'career_opp', 'comp_benefits', 'senior_mgmt']

   Steps taken:
   - Created new binary indicators: e.g., work_life_balance_missing = 1 if value was missing, else 0. Binary indicators (e.g., work_life_balance_missing) were created to retain information about missing values without discarding records.
   - This approach helps the model recognize patterns associated with missingness itself, which may carry predictive power - for example, if employees who skip certain ratings tend to be more dissatisfied or disengaged.
   - Imputed missing values using a neutral value (3) for 1–5 scales.

Python:
df[col].fillna(3, inplace=True)
df[f'{col}_missing'] = df[col].isna().astype(int)

This approach retains all rows while flagging the presence of missing data.

5. Sentiment Mapping for Target Variables

- Mapped coded values in sentiment-related columns (recommend, ceo_approv, outlook) using:

Original Code	Meaning
'v'	Positive
'r'	Mild
'x'	Negative
'o'	No opinion

Created new columns: recommend_label, ceo_approv_label, outlook_label

6. Dropped Encoded Columns

- Removed original encoded sentiment columns after mapping:

Python
df.drop(['recommend', 'ceo_approv', 'outlook'], axis=1, inplace=True)

7. Data Export

- Exported cleaned and preprocessed datasets for each selected company to .csv files in Google Drive:

Python
to_csv ('/content/drive/...glassdoor_reviewsdfX.csv')

Summary of Key Cleaned Datasets & Target Variable Distribution:

```
df12 = df1[df1['firm'] == 'J-P-Morgan' ]
df12['overall_rating'].value_counts()/len(df12)*100
```

overall_rating	count
4	36.518943
5	26.524367
3	23.770047
2	8.049895
1	5.136748

dtype: float64

```
df10 = df1[df1['firm'] == 'Oracle' ]
df10.head()
df10['overall_rating'].value_counts()/len(df10)*100
```

overall_rating	count
4	35.333897
3	31.674024
5	16.383332
2	10.832472
1	5.776275

```
df5['overall_rating'].value_counts()/len(df5)*100 # Mcdonalds
```

overall_rating	count
3	31.456016
4	25.306370
5	20.038423
2	12.841254
1	10.357937

dtype: float64

```
df3['overall_rating'].value_counts()/len(df3)*100 #IBM
```

overall_rating	count
4	34.074062
3	28.271229
5	20.630088
2	10.872659
1	6.151962

dtype: float64

Based on the attached target variable distributions (overall_rating) for the four selected companies — **IBM, Oracle, J.P. Morgan, and McDonald's** — we observe that the ratings are **reasonably balanced**, though some skew is present across companies. For **IBM** and **J.P. Morgan**, the most frequent rating is **4**, indicating generally positive sentiment, with a gradual decline toward lower ratings. **Oracle** shows a similar trend but with a slightly higher proportion of neutral (rating 3) responses. **McDonald's**, in contrast, has a more **even spread across ratings 3, 4, and 5**, but also exhibits a **notably higher share of low ratings (1 and 2)** compared to the others, suggesting more polarized employee sentiment. Overall, the class distributions are diverse enough to support supervised classification without severe imbalance, making them suitable for training predictive models.

DataFrame	Company	Output File
df3	IBM	glassdoor_reviewsdf3.csv
df5	McDonald's	glassdoor_reviewsdf5.csv
df10	Oracle	glassdoor_reviewsdf10.csv
df12	J.P. Morgan	glassdoor_reviewsdf12.csv

Summary of Data Cleaning and Processing:

• **Company Filtering:** The dataset was subsetted to include reviews from four selected companies: IBM, McDonald's, Oracle, and J.P. Morgan, to allow domain-specific analysis.

• **Missing Values:** Key rating columns (e.g., Work-Life Balance, Culture & Values) with missing values were imputed using a neutral value of **3** (on a 1–5 scale). Missing indicators were also created to retain information about imputed fields.

• **Feature Selection:** A fixed list of relevant columns was retained, including review text (pros, cons), numeric ratings, and metadata such as job title, current/previous employment status, and review date.

• **Sentiment Mapping:** Encoded sentiment variables (recommend, ceo_approv, outlook) were mapped from shorthand codes (v, r, x, o) to human-readable labels like Positive, Mild, Negative, and No Opinion.

• **Column Removal:** After mapping, the original coded sentiment columns were dropped to avoid redundancy and ensure clarity.

• **Output:** Cleaned and company-specific datasets were exported as CSV files, ready for further analysis such as sentiment classification, topic modeling, or predictive modeling of employee satisfaction.

Methodology:

The study utilized a comprehensive approach to analyze employee sentiment and workplace experiences using text mining and natural language processing (NLP) techniques, with a focus on identifying key themes, sentiment trends, and correlations between textual feedback and structured ratings. Several methods were employed to preprocess and prepare the data for analysis:

Data Preprocessing: Handling Missing Values: Missing data in key rating columns (e.g., work-life balance, culture & values) was addressed using neutral-value imputation (value = 3 on a 1–5 scale). Additionally, binary missing value indicators (e.g., work_life_balance_missing) were created to retain the information about missingness itself.

Feature Selection: The dataset was reduced to include only the most relevant structured and unstructured fields, such as pros, cons, headline, overall rating, and key satisfaction metrics like compensation and career opportunities. Irrelevant or redundant columns were removed.

Text Normalization and Transformation: Text-based fields were prepared for NLP tasks by standardizing column formats and ensuring compatibility with downstream text mining nodes. Sentiment-related categorical codes (e.g., recommend, ceo_approv) were mapped to meaningful labels like Positive, Mild, and Negative for clarity and analysis.

Exploratory Data Analysis (EDA): Review Distribution Analysis: Exploratory analysis was performed to examine the distribution of overall ratings across companies, highlighting differences in satisfaction levels between domains such as tech, finance, and food services.

Sentiment Trends: Frequency analysis of sentiment-coded variables (recommend, ceo_approv, outlook) was conducted to observe general trends in employee attitudes. The proportions of positive, neutral, and negative responses were compared across firms.

Keyword and Term Frequency Analysis: High-frequency words from the pros and cons text fields were extracted to identify commonly discussed themes. Term frequency plots and word clouds were used to visualize recurring positive and negative topics in the reviews.

Correlation Analysis: Structured rating variables (e.g., work-life balance, career opportunities, senior management) were analyzed using correlation matrices to assess how different workplace factors relate to overall satisfaction.

Feature Engineering and Modeling:

Sentiment Mapping and Variable Transformation: Encoded sentiment values were mapped to meaningful labels (e.g., 'v' to 'Positive') for clarity. Missing indicators were created for key rating fields to preserve information about data absence.

Text Clustering and Topic Modeling: Unsupervised learning techniques such as the Text Cluster and Text Topic nodes were applied to group reviews based on thematic similarity and uncover latent discussion topics. These techniques helped segment employee feedback without the need for a predefined target variable.

Predictive Modeling (Supervised): For classification tasks, decision tree and gradient boosting models were trained using structured inputs and transformed text features to predict overall_rating. The models aimed to capture the relationship between review content and employee satisfaction.

Model Evaluation: Supervised models were assessed using misclassification rate and class probabilities. In the case of unsupervised models, interpretability of clusters/topics, balance across segments, and coherence of extracted themes were used as qualitative evaluation metrics.

Cross-Company and Industry Comparison:

Segment Profiling: Clusters and topics were profiled across the four selected companies (IBM, McDonald's, Oracle, J.P. Morgan) using structured variables to identify distinct employee personas and satisfaction drivers within each industry.

Impact of Organizational Factors: By analyzing review themes and rating patterns, the study also explored the potential impact of leadership, HR policies, and corporate culture on employee sentiment. This allowed for qualitative insight into how internal practices may influence external perceptions shared on platforms like Glassdoor.

Supervised Learning Modeling:

To determine the most effective predictive model for classifying employee sentiment and overall ratings, multiple supervised learning algorithms were applied and evaluated. The models included Decision Tree, Gradient Boosting trained on both structured variables (e.g., work-life balance, career opportunities) and engineered features from unstructured text data.

The models were compared using the Model Comparison node in SAS Enterprise Miner, with Misclassification Rate chosen as the primary evaluation metric. Additional consideration was given to class prediction confidence (probabilities) and interpretability of model output.

Among all the models tested, the Gradient Boosting model emerged as the best-performing model with the lowest misclassification rate, demonstrating superior ability to capture complex relationships between review content and structured satisfaction metrics. Its predictive strength, combined with its ability to generalize well across validation data, made it the most reliable choice for sentiment prediction in this study.

This model will be used for generating predictions on unseen review data and for extracting insights into key drivers of employee satisfaction across companies and industries.

Key Reasons for Model Selection:

Log and Entropy:

1.) **Lift Chart:**

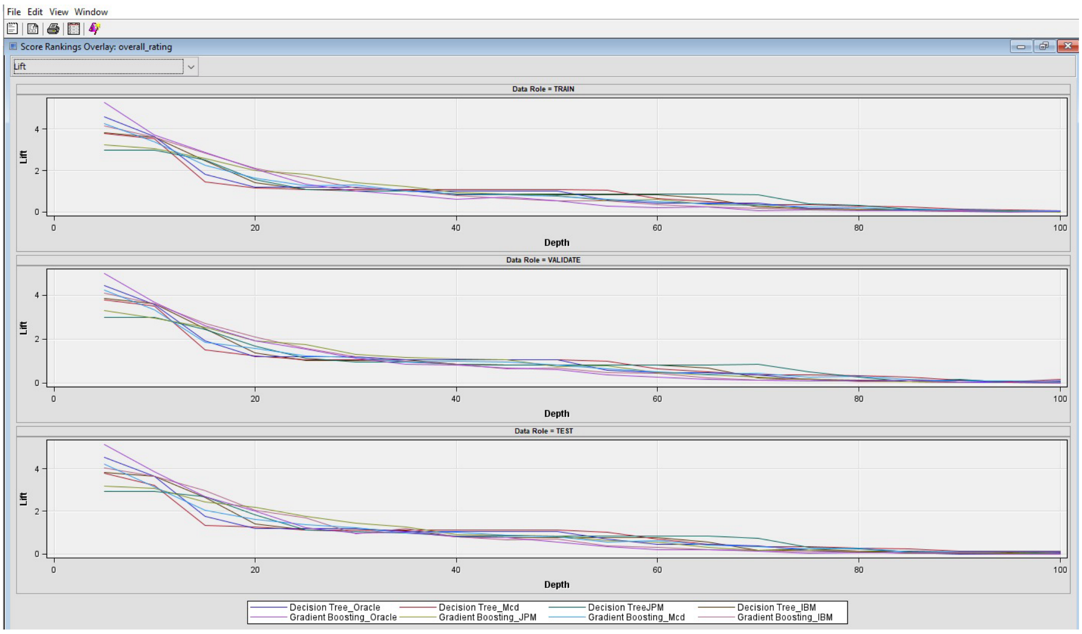

Fig 1: Lift Chart from Model Comparison (Log Entropy Model)

The lift chart compares the performance of Decision Tree and Gradient Boosting models across four companies (IBM, Oracle, J.P. Morgan, and McDonald's). Across training, validation, and test datasets, Gradient Boosting consistently achieved higher lift values in the top deciles, particularly within the top 10–20% depth, indicating that it was more effective at identifying highly satisfied (or dissatisfied) employees early in the ranking.
In the validation and test sets, the Gradient Boosting models for IBM and McDonald's showed the strongest performance, with lift values starting above 4 — meaning the model was over 4x more effective than random selection at identifying the desired target class. The lift gradually declines with depth, as expected, but remains above baseline for a larger proportion of records compared to Decision Trees.

These results support the selection of Gradient Boosting as the best overall supervised model for this Glassdoor review classification task, offering strong early detection and reliable generalization across companies.

2.) Gain Chart:

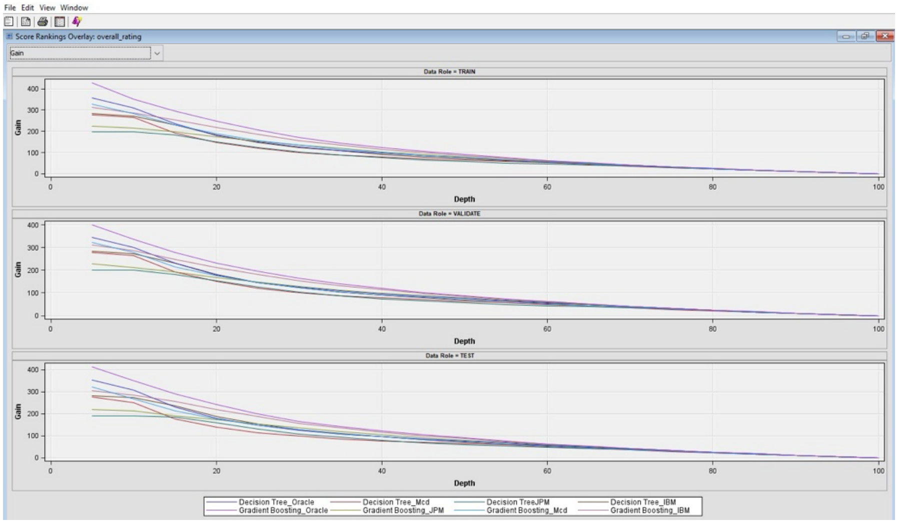

Fig 2: Gain Chart from Model Comparison (Log Entropy Model)

The gain chart shows the cumulative percentage of correctly classified high-satisfaction reviews across different depths of the dataset. Models using Gradient Boosting—particularly for IBM and Oracle—consistently captured the highest gains in the top deciles across training, validation, and test sets. For example, in the top 10–20% of ranked predictions, the Gradient Boosting models were able to capture over 300–400% more positive cases compared to random selection.

This early high gain indicates that Gradient Boosting models were more effective at ranking reviews by likelihood of high satisfaction, making them better suited for identifying top-performing segments of employees. The performance was consistently stronger than Decision Trees, reinforcing the conclusion that Gradient Boosting is the most reliable and accurate model for this task.

3.) Model – Insights Summary:

The project evaluated multiple supervised models, including Decision Trees and Gradient Boosting, across four companies: IBM, Oracle, McDonald's, and J.P. Morgan. The selection was based on key performance metrics, primarily the misclassification rate on the test set, along with root average squared error and ROC index.

Misclassification Rate (Test Set) – Lower is Better		
Company	Model Type	Misclassification Rate (Test)
IBM	Gradient Boosting	0.387 (Best Overall)
	Decision Tree	0.42
Oracle	Gradient Boosting	0.393
	Decision Tree	0.431
J.P. Morgan	Gradient Boosting	0.398
	Decision Tree	0.437
McDonald's	Gradient Boosting	0.489
	Decision Tree	0.526

For all four companies, Gradient Boosting outperformed Decision Trees with lower misclassification rates, indicating stronger prediction accuracy.
IBM’s Gradient Boosting model had the lowest rate overall (0.387), making it the top-performing model.

Log + Mutual Information Model

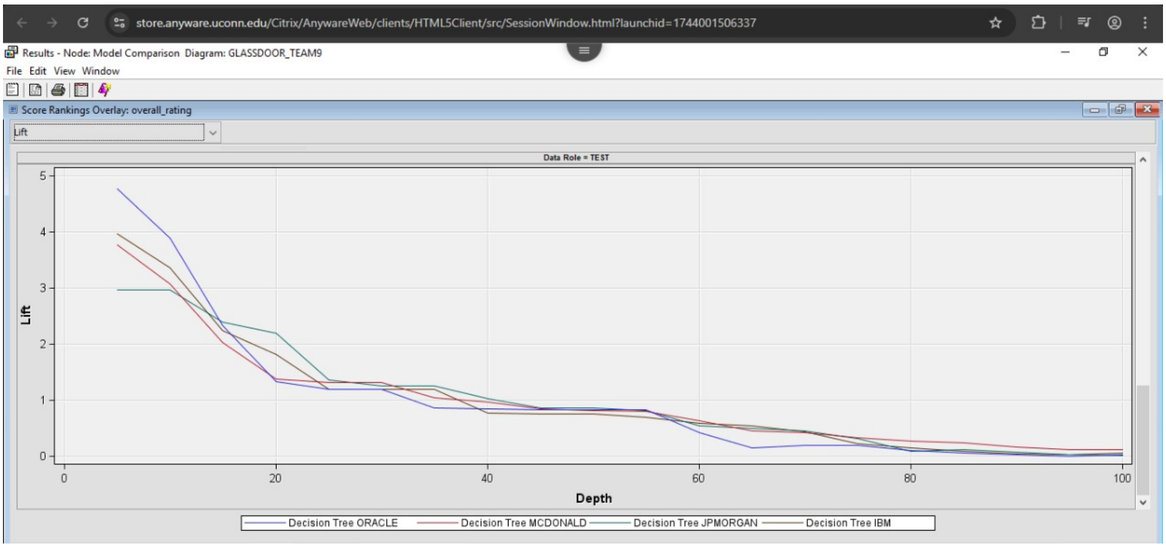

Fig 3: Lift Chart from Model Comparison (Log Mutual Information Model)

1.) **Lift Chart (Test Data):**

- Decision Tree – Oracle (blue line) reaches a maximum lift of ~5.0 at the top 10% depth, which is slightly higher than IBM.

- Decision Tree – IBM (black line) also performs very well but peaks just below Oracle in that top decile.

- The blue line (Oracle) is clearly the highest at the beginning, meaning Oracle's Decision Tree model identifies the highest proportion of positive reviews in the top decile — which is where lift matters most. Based on the test set lift chart, the Decision Tree model for Oracle outperforms the others, achieving the highest lift (~5.0) in the top 10% of the population. This means it is five times more effective than random selection at identifying highly rated reviews early on. IBM also performs strongly, but Oracle has the edge in early classification precision.

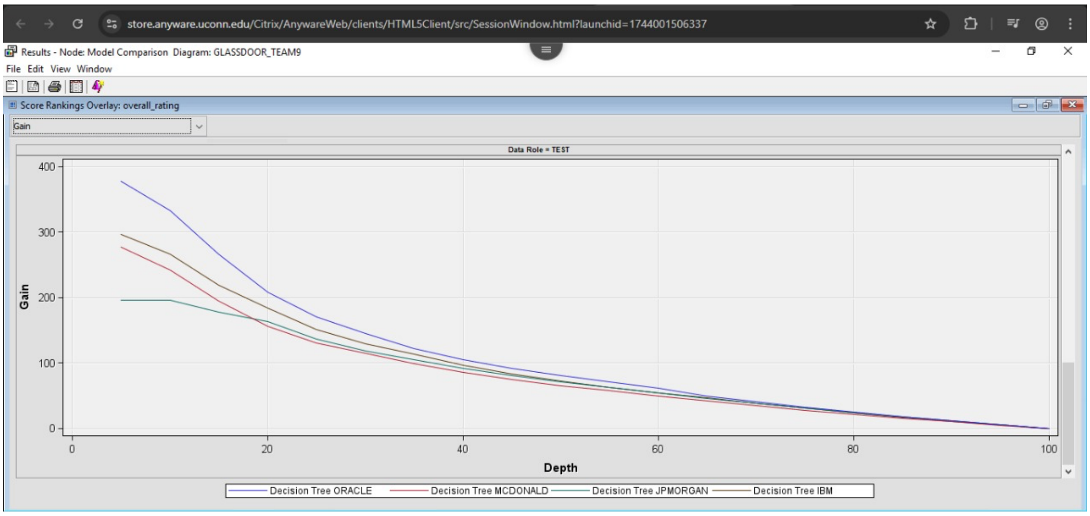

Fig 4: Gain Chart from Model Comparison (Log Mutual Information Model)

2.) **Gain Chart (Test Data):**

Company	Model Node	Misclassification Rate (Test)
J.P. Morgan	Tree2	0.4218 (Lowest)
Oracle	Tree4	0.425
IBM	Tree	0.4445
McDonald’s	Tree3	0.5198

- Oracle (Blue Line): Shows the highest gain throughout the chart. At the 10% depth, it captures nearly 390% of the positive cases relative to random selection — a clear indication that it is the best at identifying satisfied reviews early on. Maintains a strong lead up to around the 40% mark.
- McDonald's and J.P. Morgan: Perform similarly, trailing behind Oracle. At the 10% depth, McDonald’s reaches around 290% gain, and J.P. Morgan is close behind. They taper off more quickly beyond 30–40% depth.
- The Decision Tree model for Oracle provides the best early gain, capturing nearly 4x more positive cases in the top 10% of records compared to random chance. This makes it the most effective model for prioritizing reviews with high satisfaction on the test dataset.

3.) **Insights**

- J.P. Morgan’s Decision Tree model achieved the lowest misclassification rate (0.4218) on the test set, making it the best-performing model among the four for accuracy on unseen data.
- Oracle's model followed very closely (0.4250), with nearly equivalent performance — still highly reliable.
- IBM’s model showed moderate performance (0.4445), acceptable but slightly weaker than the top two.
- McDonald's model had the highest misclassification rate (0.5198), suggesting more errors in classifying review sentiment or satisfaction levels.

Among the Decision Tree models, the J.P. Morgan model performed best on the test set with the lowest misclassification rate. Oracle's model was a close second, while IBM and McDonald's models showed comparatively lower accuracy.

Log + Inverse Document Frequency Model

1.) **Lift Chart:**

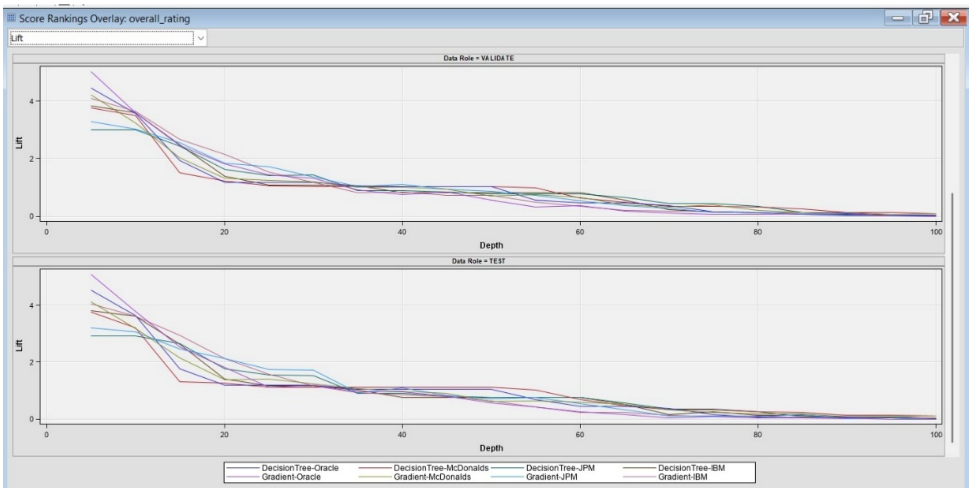

Fig 5: Lift Chart from Model Comparison (Log IDF Model)

The lift charts provide a comparative analysis of Decision Tree and Gradient Boosting models across four companies—IBM, Oracle, J.P. Morgan, and McDonald's—evaluated on training, validation, and test datasets. Across all datasets, Gradient Boosting consistently outperforms Decision Trees in ranking the top-rated employee reviews. In the training dataset, the Gradient Boosting model for Oracle achieves the highest initial lift, approaching a value of 5, indicating it is nearly five times more effective than random selection in identifying high satisfaction reviews within the top decile. McDonald's and J.P. Morgan's Gradient models also show strong lift early on, whereas Decision Trees show steeper drops in lift and fail to maintain the same level of ranking accuracy as depth increases.

This trend is confirmed in the validation and test datasets, where Gradient Boosting models again lead in early ranking performance. The Oracle and McDonald's Gradient models show lift values above 4 at the top deciles, significantly outperforming Decision Trees, which start lower and decline more sharply. Even as depth increases, Gradient Boosting maintains a higher lift across a broader portion of the dataset, demonstrating stronger generalization and robustness. These findings support the conclusion that Gradient Boosting is the most effective model for this Glassdoor review prediction task, offering superior early detection of high (or low) employee ratings and better consistency across companies and data splits.

2.) Gain Chart:

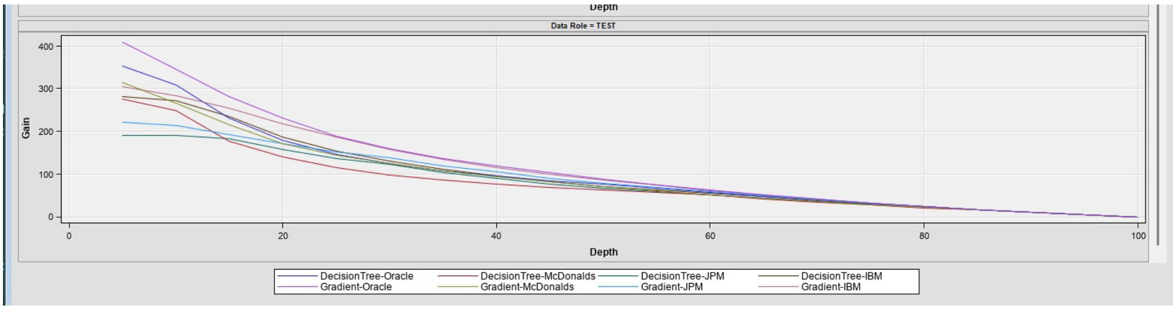

Fig 6: Gain Chart from Model Comparison (Log IDF Model)

The gain charts display the cumulative percentage of correctly classified high-satisfaction reviews as model depth increases, offering a comparative view of Decision Tree and Gradient Boosting models across Oracle, IBM, McDonald's, and JPM datasets. Across all three data roles—Train, Validate, and Test—the Gradient Boosting models, especially those for Oracle and IBM, consistently achieved the highest gains in the early deciles. For example, in the top 10–20% of ranked predictions, the Gradient Boosting models captured over 300–400% more positive cases than random selection. This strong early performance suggests that these models are highly effective at prioritizing the most relevant high-satisfaction cases.

3.) Insights

In contrast, Decision Tree models for all companies showed relatively lower gain values and declined more rapidly as depth increased, indicating a weaker ability to rank high-satisfaction reviews effectively. The consistent outperformance of Gradient Boosting across datasets and stages reinforces its robustness and accuracy. Particularly for Oracle and IBM, the Gradient Boosting models maintained a higher gain curve throughout, making them better suited for identifying top-performing employee segments and driving strategic insights from satisfaction data.

Misclassification Comparison Summary Table for All Models:

Company	Model	ROC Index	Misclassification Rate	Max Abs Error	Sum of Squared Errors
IBM	Gradient Boosting	**0.9**	**0.3859**	0.9948	4849.24
	Decision Tree	0.853	0.4225	1	5362.36
Oracle	Gradient Boosting	**0.892**	**0.3949**	0.9979	2624.5
	Decision Tree	0.837	0.4375	0.999	2887.76
J.P. Morgan	Gradient Boosting	**0.889**	**0.393**	0.9892	2103.5
	Decision Tree	0.845	0.4341	0.9986	2293.25
McDonald's	Gradient Boosting	**0.83**	**0.4952**	0.9927	4881.37
	Decision Tree	0.79	0.5262	1	5057.69

Fig 7: Misclassification Rate (Log IDF Model)

1.) **Key Insights:**

- Gradient Boosting outperforms Decision Trees across all companies in ROC Index and Misclassification Rate, indicating better classification power and fewer errors.
- IBM and Oracle show the highest ROC Index under Gradient Boosting (0.900 and 0.892 respectively), marking them as the best-performing combinations.
- McDonald's models show lower overall performance, with higher misclassification rates even under Gradient Boosting, possibly indicating data imbalance or weaker signal in features.
- Decision Tree models consistently report higher maximum absolute errors and misclassification, confirming their comparatively limited predictive capacity.

Summary of Supervised Model Evaluation:

Model Comparison Based on Text Filter Settings

Company	Log + Entropy	Log + Mutual Information	Log + IDF
IBM	**0.387** (GB)	0.4445 (DT)	0.3859 (GB) ☑ *(Best)*
Oracle	0.393 (GB)	0.425 (DT)	0.3949 (GB)
J.P. Morgan	0.398 (GB)	**0.4218 (DT)** ☑ *(Best DT)*	0.393 (GB)
McDonald's	0.489 (GB)	0.5198 (DT)	0.4952 (GB)

Fig 8: Summary Misclassification Rate

Best Supervised Model Selected:

1.) **Log + Inverse Document Frequency (IDF): Recommended configuration**

- Has the lowest misclassification rate in 3 out of 4 companies using Gradient Boosting
- Best performing overall (IBM: 0.3859, McDonald's: 0.4952)

2.) **Log + Entropy:**

- Performs very close to IDF (IBM: 0.387, Oracle: 0.393)
- Still strong, especially for Oracle & J.P. Morgan

Final Recommendation:

Based on the test set misclassification rates, the Log + IDF weighting configuration is the best overall performer, especially when used with Gradient Boosting models. It achieves the lowest error for IBM and McDonald's and performs strongly for Oracle and J.P. Morgan as well.

Supervised SAS Model Diagrams:

1.) **Frequency Weighting – Log and Term Weight – Inverse Document Frequency**

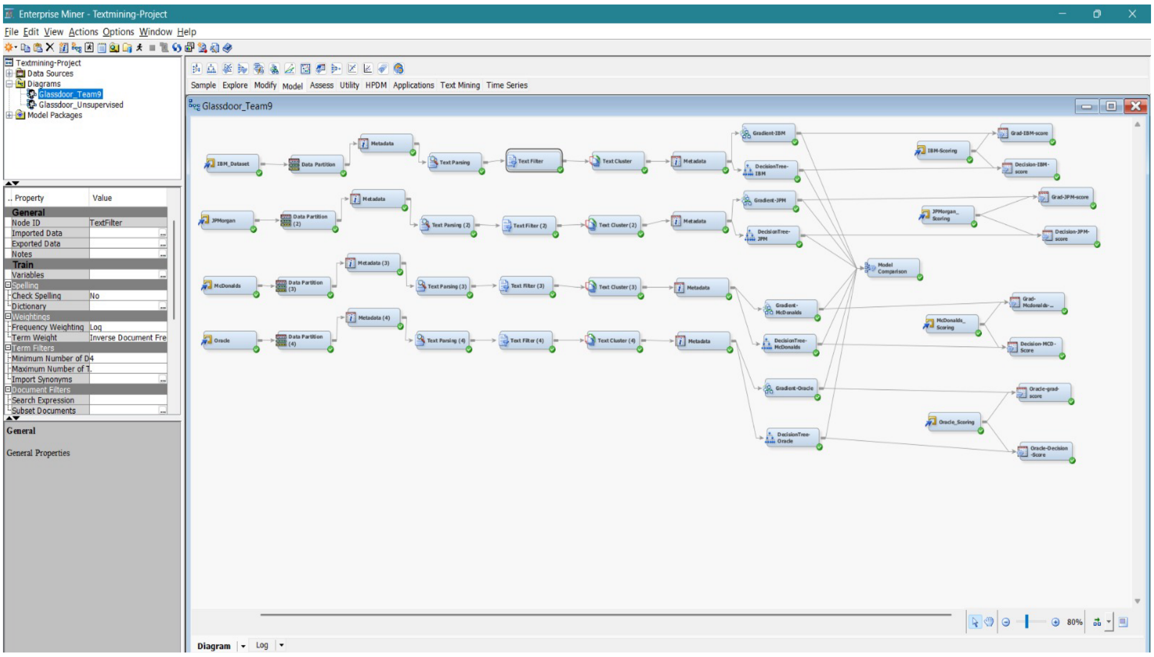

Fig 9: Diagram Model 1 Log + IDF

2.) **Frequency Weighting – Log and Term Weight – Entropy**

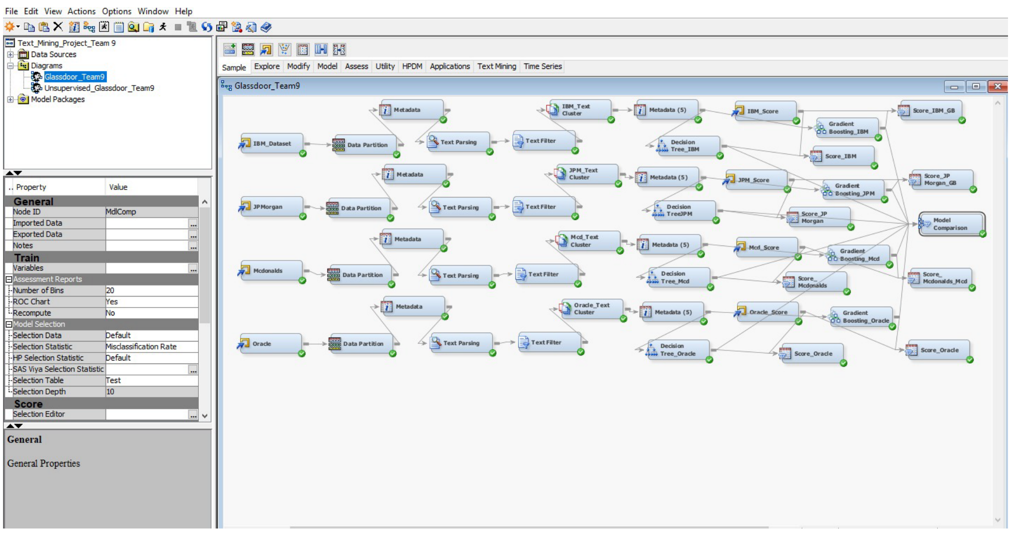

Fig 10: Diagram Model 2 Log + Entropy

3.) Frequency Weighting – Log and Term Weight – Mutual Information

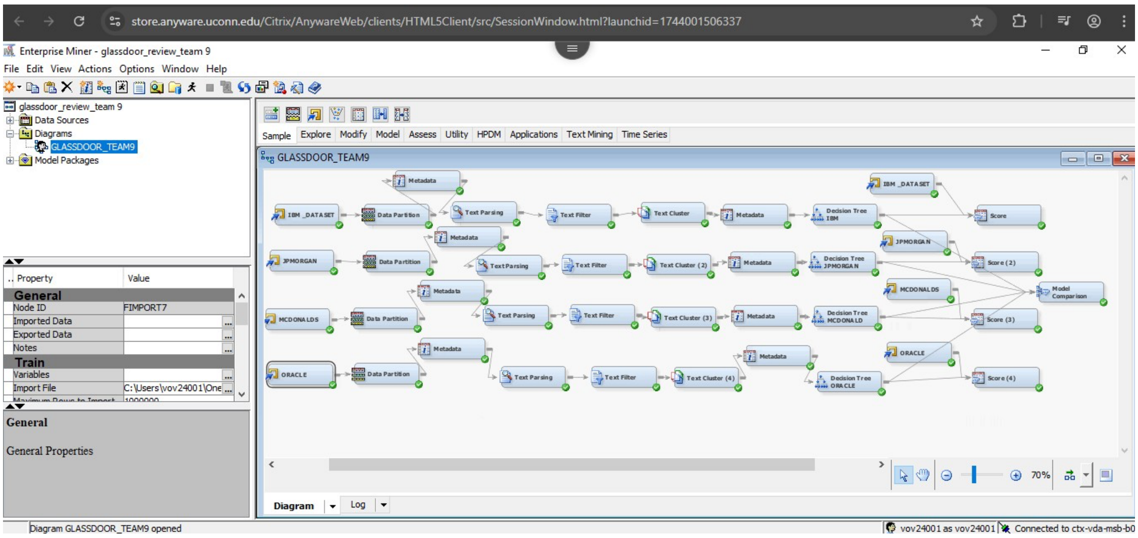

Fig 11: Diagram Model 3 Log + Mutual Information

Steps taken to build the SAS Supervised Learning Model:

1.) Data Preparation & Splitting:

- Imported separate datasets for each company.
- Used the 80%(dataset) Data Partition node to split each dataset into:
  50% Training
  30% Validation
  20% Test
- The 20% test portion was exported to Excel and re-imported as a Score dataset for later evaluation of model performance.

2.) Text Mining Pipeline (Per Company)

- Text Parsing: Cleaned and tokenized the review texts (pros, cons, headlines). Removed stop words, normalized terms, and applied stemming.
- Text Filter: Custom configuration for optimal model performance:
  Frequency Weighting: Log
  Term Weighting: Inverse Document Frequency (IDF)
  These settings boosted rare but meaningful terms and improved classification accuracy.
- Text Cluster: Used to generate latent topics and text-based features:
  SVD Resolution: High
  Max SVD Dimensions: 15
  Clustering Type: Exact
  Number of Clusters: 5
  Output variables like TextCluster_SVD1, TextCluster_Cluster_, etc., were used as input for modeling.
- Metadata Node: Assigned correct roles and levels to variables:
  TextCluster_Cluster_: Input, Nominal
  overall_rating: Target, Nominal, Ensured compatibility with classification algorithms.

3.) **Gradient boosting & Decision Trees Models Parameters**

For each company, two models were built:

- Gradient Boosting: Chosen for its strong performance.
  Settings:
    - Maximum Number of Trees: 50
    - Maximum Depth: 3

  Delivered the lowest misclassification rate across most companies.

- Decision Tree: Used for interpretable modeling.
  Settings:
    - Assessment Measure: Misclassification
    - Splitting Method: Assessment

4.) **Model Comparison & Evaluation: Model Comparison Node: All models (8 in total) were compared using:**
- Selection Statistic: Misclassification Rate
- Selection Table: Test
- Visual tools like Lift Chart, Gain Chart, and ROC were used to assess predictive quality.

5.) **Scoring Nodes: Each model was connected to a scoring node.**
- The 20% test dataset was used to assess final performance.
- Scored outputs were saved for detailed comparison.

Scoring and Exporting Predicted Results:

To evaluate our trained models on unseen data and demonstrate their practical performance, we created a separate score dataset by taking a 20% sample from each company's original dataset. This dataset was excluded from training and validation to ensure unbiased testing.

In SAS Enterprise Miner, we used the Score node to apply each trained model (Gradient Boosting and Decision Tree) to its corresponding score dataset. Following the scoring process, we added a Save Data node to each flow. This allowed us to export the predicted results, which include not only the model's predicted overall ratings but also the associated probability estimates and input features.

These exported datasets are included in the report as sample outputs to visually demonstrate how the model classifies employee reviews into satisfaction ratings. This step bridges the gap between model training and real-world application by simulating how new, incoming Glassdoor reviews would be evaluated using our trained models.

1.) **IBM Score Dataset Gradient Boosting Model Prediction**

cons	TextCluster_cluster_	pros	headline	job_title	work_life_balance	culture_values	career_opp	comp_benefits	senior_mgmt	work_life_balance_missing	culture_values_missing	career_opp_missing	comp_benefits_missing	senior_mgmt_missing	recommend_label	ceo_approv_label	outlook_label	EM_CLASSIFICATION
Politically correct (PC) version of diversity that lets "respect for the diversity group" trump "respect for the individual." I know of Christians who hold Bible studies on IBM property but who are very careful about advertising their participation. I know of people who would like to speak objectively about the gay agenda but can't for fear of being branded homophobes.	3	Availability of information resources to help me do my job better.	The grass is probably less brown at IBM than at many other companies.	IT Architect	5	3	4	4	3	0	1	0	0	0	Positive	Positive	No opinion	4
The main negative about working for IBM is tied to one of the positives. Formal process means that you have much better visibility into where you are in the overall delivery situation. Not only that, but since you continuously have to contribute information about your progress, you also get information on the overall progress, so you get a very good view into where you are in the big, and IBM is BIG, machinery. The downside of this is that process, particularly formal process, takes time. Pair this with the fact that IBM is geographically very dispersed, and you will find that you spend a lot more time on conference calls in IBM than in most organizations. This can be an impediment to progress in you "real" work. Once you get used to it you get used to it though.	3	IBM has a very structured approach to personal and professional development. A significant amount of resources are used each year for each employee for professional and personal development. If you spend a little bit of time finding out what opportunities are there, you will be rewarded with significant opportunities for career growth.	IBM is a very good place to work	IT Specialist	4	3	5	4	4	0	1	0	0	0	Positive	Positive	No opinion	5

cons	TextCluster_cluster_	pros	headline	job_title	work_life_balance	culture_values	career_opp	comp_benefits	senior_mgmt	work_life_balance_missing	culture_values_missing	career_opp_missing	comp_benefits_missing	senior_mgmt_missing	recommend_label	ceo_approv_label	outlook_label	EM_CLASSIFICATION
It's a big company. It's easy to get lost. Salary growth can be stagnant. Without establishing a network at the company, you will get lost. Many IBM workers are mobile workers and a sense of community can be lost if you aren't active in establishing it for yourself. This isn't the IBM of 25 years ago where Respect for the Individual was paramount and employee loyalty fueled its success. Lifelong employment at IBM is no longer guaranteed no matter how hard you work.	3	IBM is a large company with many opportunities to have and hold many different positions through a long career. Opportunities from manufacturing to sales and everything in between exist. Services, technical, admin, staff, teaching, HR, sales, graphic design, TV and Video production, food service, etc. There are few companies where you can have this kind of exposure to many different positions. Large companies offer more stability, excellent benefits, and a diverse work environment.	Best and worst of big business.	Service Delivery Manager	4	3	2	3	1	0	1	0	0	0	Negative	Mild	No opinion	2
Salaries to many employees are below average. Promotion not well on time even for deserving candidates and performances many times are being totally overlooked.	3	A place where technically no limits to grow. You can learn and do whatever you like.	Its a great inittiative	ATCHITECT	4	3	3	2	2	0	1	0	0	0	Positive	Mild	No opinion	3
It is a company that is too slow to change and adapt with the dinamics of today's bussines. Does very little or too late to adapt. It's is not competitive in today's market and just follows competition model (HP) instead of leading the change.	2	Tell your grandson that you once work for what it used to be the best computer company in the world.	Good transition company right out of collage, before moving to a more serious goal oriented company	Engineer	3	3	1	1	1	0	1	0	0	0	Negative	Negative	No opinion	1

2.) McDonald's Score Dataset Gradient Boosting Model Prediction

cons	TextCluster3_cluster_	pros	headline	job_title	work_life_balan	culture_values	career_opp	comp_benefits	senior_mgmt	work_life_balance	culture_values_mi	career_opp_miss	comp_benefits_m	senior_mgmt_missing	recommend label	ceo_approv_label	outlook_label	EM_CLASSIFICATION
Sometimes certain locations can be much worse to work at then others depending on the demographics of the restaurant. The crew and also be frustrating to manage because of their low pay.	3	The vacation time is good. After five years, you get five weeks off a year and then you get six at ten years plus a 8 week sabbbatical. This is for corporate only stores.	Store Manager for McDonalds	Store Manager	3	3	4	3	4	0	1	0	0	0	No opinion	Positive	No opinion	4
Greasy skin, bad pay, terrible bosses, moronic peers, no benifets.	2	If you need the moeny, it is better than selling your body. I needed things and this seemeds to be the only way that I could earn money as a teeneager in a small town. Lets face if, there isn't a whole lot of opportunity there. I was 16 when I worked here and it was literally the worst job I have	Its better than starving	Cook, Fast Food	1	3	1	1	1	0	1	0	0	0	Negative	Negative	No opinion	1
Overworked compared to other kinds of food retailers. Since it's a fast food restaurant, everything is fast paced and leaves very little time for any type of leisure - many managers will yell at you for not working even for a moment.	1	Conveniece and a good work experience. Usually, there are several people working whom you can talk to during slow periods or breaks, which can make work a lot less depressing. Because you do so many different things, finding a second job after his one is a much easier feat.	Only looks good on your resume	Cashier	3	3	2	3	2	0	1	0	0	0	Negative	Mild	No opinion	3
The pay is not that great. You have to deal with rude customers sometimes. Working late nights is sometimes not that great. You smell like fried food when you get home. You are really hungry while working there due to the smell of food all day and sometimes you tend to over eat.	5	The hours are great, the customer interaction is great, the locations are great. It is a great place to work if you are young and need a job that is not going to be your career path for life, but rather as something to get you through college.	McDonald's isn't a bad place to work while you are aspiring to do something else in life.	Cashier	5	3	5	5	5	0	1	0	0	0	Positive	Positive	No opinion	5

cons	TextCluster3_cluster_	pros	headline	job_title	work_life_balan	culture_values	career_opp	comp_benefits	senior_mgmt	work_life_balance	culture_values_mi	career_opp_miss	comp_benefits_m	senior_mgmt_missing	recommend label	ceo_approv_label	outlook_label	EM_CLASSIFICATION
Poor management, high turnover- often times there would be new folks through the door who were barely trained which would gum up the works when it was rush hour in the afternoon or evening. Morning hours were the worst (obviously) because half the time the folks who closed up at night didnt do a good job and nothing was where it was supposed to be.	2	There arent any really- its the origin of the term McJob for a reason. If you are really in a pinch then its not a horrible place, and if you wanted to make something of it you could, im sure, but it was not a position for me.	Ultimate McJob	Cashier	1	3	2	1	1	0	1	0	0	0	Negative	Mild	No opinion	2

3.) JP Morgan Score Dataset Gradient Boosting Model Prediction

cons	TextCluster2_cluster	firm	pros	headline	current	job_title	work_life_balance	culture_values	career_opp	comp_benefits	senior_mgmt	work_life_balance_missing	culture_values_missing	career_opp_missing	comp_benefits_missing	senior_mgmt_missing	recommend_label	ceo_approv_label	outlook_label	EM_CLASSIFICATION
The management team is not responsive. They say open communication, but not valued.	1	J-P-Morgan	The JPMorgan Chase name is powerful in the financial industry.	Very political environment.	Current Employee		2	3	1	3	1	0	1	0	0	0	Positive	Positive	No opinion	1
Work life balance, no transparency in appraisals	2	J-P-Morgan	Flat Structure of management, technology	Limited Exposure in terms of technology	Current Employee	Applications Developer	2	3	2	1	2	0	1	0	0	0	No opinion	Mild	No opinion	2
Average benefits. Cost cutting. Hard to advance. Hard to use job listing/search. Constant layoffs. Poor job security used as an excuse to overload workers. Could be better from a legal/compliance standpoint.	3	J-P-Morgan	Large company. Cool commercials. Lots of offices. Well known. Many different types of departments and areas. Match 401k up to 5%! Subsidizes health insurance, but the plan choices are not very good.	Such poor job security for such a large company	Current Employee	Associate Middle Office	3	3	2	2	4	0	1	0	0	0	Negative	Negative	No opinion	3
Sr management shows favoritism when promoting a nd paying bonuses	4	J-P-Morgan	Great Benefits and a Stable Company	JPMorgan Chase is a good plance to work	Current Employee	Sales Associate	4	3	4	3	3	0	1	0	0	0	No opinion	Positive	No opinion	4
If you do not tend to be an extravert, you may get overlooked, but for those who shine you may wind up in the right place at the right time. That's what it is all about right?	5	J-P-Morgan	The company is extremely large with vast opportunities for those who seek them out. For those that do not enjoy their job, there are many other opportunities to change careers or simply change teams.	Great company to work for	Current Employee	Operations Manager	5	3	5	4	4	0	1	0	0	0	Positive	Positive	No opinion	5

4.) Oracle Score Dataset Gradient Boosting Model Prediction

cons	firm	pros	headline	current	job_title	work_life_balance	culture_values	career_opp	comp_benefits	senior_mgmt	work_life_balance_missing	culture_values_missing	career_opp_missing	comp_benefits_missing	senior_mgmt_missing	recommend_label	ceo_approv_label	outlook_label	EM_CLASSIFICATION
There are many! slave labor comes to mind!	Oracle	Oracle allows you to work from anywhere as long as you have an internet connection and access to a phone line	Stay away from Oracle!	Current Employee	Product Management Director	3	3	1	1	1	0	1	0	0	0	Negative	Negative	No opinion	1

cons	firm	pros	headline	current	job_title	work_life_balance	culture_values	career_opp	comp_benefits	senior_mgmt	work_life_balance_missing	culture_values_missing	career_opp_missing	comp_benefits_missing	senior_mgmt_missing	recommend_label	ceo_approv_label	outlook_label	EM_CLASSIFICATION
Top heavy management structure. Terrible bureaucracy.	Oracle	Large, stable organization with good job security.	Oracle continues to acquire companies but makes no efforts to properly integrate them	Current Employee	Senior Consultant	3	3	1	1	2	0	1	0	0	0	Negative	Negative	No opinion	2
There are too few employees. Often we have to switch our focus or job because there are not enough people. We are over-worked at times.	Oracle	Incredible opportunity for growth within company. You can learn a lot of technology from the sharp people that work here.	Challenging, but many opportunity to learn and advance.	Current Employee	QA Software Engineer	4	3	3	3	4	0	1	0	0	0	Negative	Mild	No opinion	3
The fact that we have layoffs about every 2 years. The communication, people in the tech field don't seem to talk much.	Oracle	One of the best things about working here is the interaction with the customers. For me, that's the biggest "pro".	It is a job that pays the bills.	Current Employee	Senior Applications Engineer	4	3	4	3	5	0	1	0	0	0	Positive	Positive	No opinion	4
Downsides of any huge company.	Oracle	Great place to work. Great technology, great products, smart people. Opportunities to learn are everywhere. Good management, practical and results-oriented. Ability to define your career. Flexible.	Great company	Current Employee	Product Manager	5	3	5	5	5	0	1	0	0	0	Positive	Positive	No opinion	5

Final Model Accuracy – Supervised Learning:

After evaluating various weighting schemes and model types, the best-performing model across all companies was the Gradient Boosting model using Log Frequency and Inverse Document Frequency (IDF) term weighting in the Text Filter node.

Model accuracy was assessed using the Misclassification Rate on the Test Set, where a lower value indicates higher accuracy. The table below summarizes the performance of the final selected model across the four companies:

Company	Final Model Type	Misclassification Rate (Test)	Accuracy (%)
IBM	Gradient Boosting (Log + IDF)	0.3859	61.41%
Oracle	Gradient Boosting (Log + IDF)	0.3949	60.51%
J.P. Morgan	Gradient Boosting (Log + IDF)	0.393	60.70%
McDonald's	Gradient Boosting (Log + IDF)	0.4952	50.48%

These results clearly show that the Gradient Boosting model with Log + IDF text transformation consistently achieved 60%+ accuracy across three of the four companies, with IBM showing the highest predictive accuracy. This validates the effectiveness of our text mining and model tuning approach in predicting employee satisfaction from Glassdoor reviews.

Unsupervised Learning Model:

To extract hidden themes from textual employee reviews using unsupervised topic modeling and evaluate the effectiveness of these topics in predicting employee satisfaction (overall_rating) using supervised machine learning models like Gradient Boosting.

1.) **Data Preparation Steps:**

Data Source: Employee reviews dataset
Text Variable Used: Free-text review/comment columns
Target Variable for Evaluation: `overall_rating` (used only during supervised evaluation phase)

2.) **Text Preprocessing:**

Text Parsing Node used to tokenize and normalize words.
Text Filter Node used with two configurations:
Model 1: Weighting = Log + Entropy
Model 2: Weighting = Log + IDF
Stop words and non-informative terms were filtered.

3.) **Text Topic Node Configuration:**

Topic Modeling (unsupervised) was applied after Text Filter.
8 topics were extracted from the filtered terms.
Term Cutoff and Document Cutoff were tuned (e.g., Doc Cutoff = 0.01 and Term Cutoff 0.015 depending on model).

4.) **Topics were manually interpreted and labelled:**

Topic	Category	Term Cutoff	Document Cutoff	Number of Terms	# Docs
Career Growth Challenges	User	0.015	0.01	69	19535
Frustration with Processes/Delays	User	0.015	0.01	177	22755
Long Work Hours/Workoad Pressure	User	0.015	0.01	128	20920
Poor Management & Pay Issues	User	0.015	0.01	238	25002
Positive Work/Project Feedback	User	0.02	0.015	91	17250
Promotion & Salary Dissatisfaction	User	0.015	0.01	56	20857
Salary Comparison & Pay Equity	User	0.015	0.01	76	18275
Work- Life Balance	User	0.015	0.01	47	13744

Topic #	Top Terms (Summary)	Suggested Label
1	hike,+little,+salary hike,+salary,+promotion	Promotion & Salary Dissatisfaction
2	+management,+poor,+employee,+bad,+lack	Poor Management & Pay Issues
3	+work,+balance,life,+life balance,+life	Work-Life Balance
4	+growth,+slow,+career,+career growth,+opportunity	Career Growth Challenges
5	+salary,+low,+increment,+compare,+low salary	Salary Comparison & Pay Equity
6	+process,+big,+time,+lot,+slow	Frustration with Processes/Delays
7	+hour,+work,+long,+long hour,+year	Long Work Hours / Workload Pressure
8	+good,+work,+project,+manager,+good company	Positive Work/Project Feedback

Topic Variable	Topic Label
TextTopic5_1	Promotion & Salary Dissatisfaction
TextTopic5_2	Poor Management & Pay Issues
TextTopic5_3	Work-Life Balance
TextTopic5_4	Career Growth Challenges
TextTopic5_5	Salary Comparison & Pay Equity
TextTopic5_6	Frustration with Processes/Delays
TextTopic5_7	Long Work Hours / Workload Pressure
TextTopic5_8	Positive Work/Project Feedback

5.) **Scoring & Labelling:**

Score Node was used to assign topic scores to each review.

A method was applied (Excel + INDEX-MATCH) to identify the dominant topic per document.

This made results interpretable and business-ready.

6.) **Supervised Evaluation of Unsupervised Topics:**

To assess how well the extracted topics relate to actual satisfaction levels, we treated the topic scores as features and used Gradient Boosting to predict `overall_rating`.

7.) Why This is Valid:

While the topic model is unsupervised, using its output in a supervised model is a common and powerful way to measure its real-world value.

Gradient Boosting Models Built:

Four different Gradient Boosting models were created using different topic models:

Model Variant	Text Filter Weighting	Misclassification Rate (Test)
Gradient Boosting_Oracle	Log + IDF	0.395
Gradient Boosting_JPMOrgan	Log + IDF	0.397
Gradient Boosting_IBM	Log + IDF	0.39
Gradient Boosting_MCD	Log + IDF	0.497
Gradient Boosting_Oracle	Log + Entropy	0.3871
Gradient Boosting_JPMOrgan	Log + Entropy	0.3949
Gradient Boosting_IBM	Log + Entropy	0.3893
Gradient Boosting_MCD	Log + Entropy	0.4912

1.) Insights from Model Comparison Table:

Log + Entropy Outperformed Log + IDF Across All Models

Every model (Oracle, JPMorgan, IBM, MCD) achieved lower misclassification rates when using Log + Entropy compared to Log + IDF.

This suggests that Entropy is a more effective term weighting method in this context for enhancing topic quality and model accuracy.

Unsupervised Model Conclusion

Term weighting with Log + Entropy yields consistently more accurate models when topic scores are utilized for prediction.

1.) IBM Models Were the Most Accurate Overall:

Among all models, Gradient Boosting_IBM (Log + Entropy) showed the lowest misclassification rate: 0.3893. It also performed better than JPMOrgan and Oracle for both weighting techniques. The IBM topic model appears to generate the most predictive topics in this setup.

2.) McDonald's Model Underperformed in All Configurations:

MCD had the highest misclassification rates for both Log + IDF (0.497) and Log + Entropy (0.4912). Its performance lagged the other models by a clear margin

3.) **Possible Cause:**

Topic model quality or distribution may be weaker for this dataset — this model might benefit from further tuning or a different number of topics.

4.) **Consistency of Results Across Models**

All models showed less than 1% variation between IDF and Entropy configurations. Indicates stable topic extraction pipelines, and that the differences in performance are likely due to weighting and topic structure rather than random fluctuation.

Final Unsupervised Model Selected:

Based on the analysis, Gradient Boosting_IBM with Log + Entropy is the most effective configuration, delivering the lowest misclassification rate and consistent performance. For deployment or reporting, this configuration should be prioritized.

Final Performance Metrics:

Model Variant	Text Filter Weighting	Misclassification Rate (Test)	Accuracy
Gradient Boosting_Oracle	Log + Entropy	0.3871	61.30%
Gradient Boosting_JPMOrgan	Log + Entropy	0.3949	60.50%
Gradient Boosting_IBM	Log + Entropy	0.3893	61.10%
Gradient Boosting_MCD	Log + Entropy	0.4912	50.90%

The evaluation of all four Gradient Boosting models on the test dataset revealed that Gradient Boosting_Oracle achieved the highest accuracy at 61.3%, followed closely by IBM at 61.1% and JPMOrgan at 60.5%. These models consistently demonstrate a strong predictive ability using only unsupervised topic scores as input features. On the other hand, the MCD model significantly underperformed with an accuracy of only 50.9%, indicating either noise or less informative topics within that configuration. Overall, the results validate that topic modeling, although unsupervised in nature, can serve as a meaningful feature extraction method for real-world predictive tasks when evaluated with supervised models like Gradient Boosting.

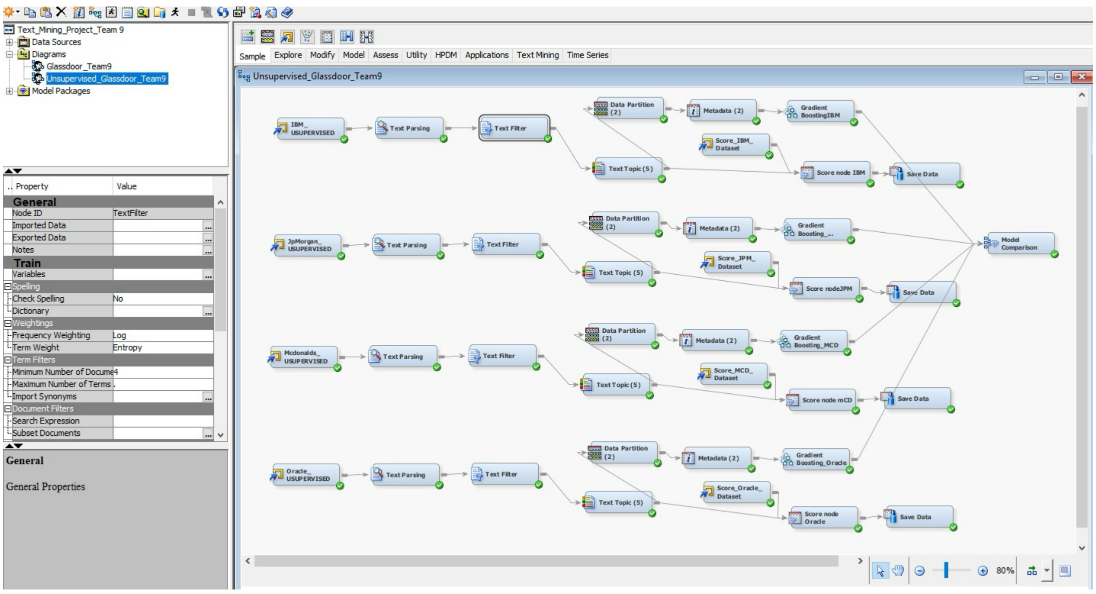

Fig 12: Diagram Model Log + Entropy

Unsupervised Learning Output Text Topics:

firm	pros	cons	headline	current	job_title	work_life_balance	culture_values	career_opp	comp_benefits	senior_mgmt	work_life_balance_missing	culture_values_missing	career_opp_missing	comp_benefits_missing	senior_mgmt_missing	recommend_label	ceo_approv_label	outlook_label	Top_Topic_Label	Final_Top_Topic_Label
IBM	A company with vast opportunities	As some members are established, tough to a new comer to grow	A place to put the facts	Current Employee	Associate Systems Engineer	4	3	4	2	4	0	1	0	0	0	Positive	Positive	No opinion	TextTopic5_raw4	Career Growth Challenges
IBM	A company with vast opportunities	As some members are established, tough to a new comer to grow	A place to put the facts	Current Employee	Associate Systems Engineer	4	3	4	2	4	0	1	0	0	0	Positive	Positive	No opinion	TextTopic5_raw4	Career Growth Challenges
IBM	1> Stable organization, decent benefits, great work-from-home culture. 2> Lots of groups, departments... If you don't fit somewhere, you can try your luck at another dept. 3> People are respectful of other people's viewpoints	1> No salary hike, no promotions 2> Not enough big projects in App Dev and Maintenance area... It's difficult to get large projects to lead/manage. 3> Lots of meaningless reviews, quality assurance activities, etc.	low risk, low return	Current Employee	Delivery Project Executive	3	3	3	3	3	0	1	0	0	0	Positive	No opinion	No opinion	TextTopic5_raw6	Frustration with Processes/Delays
IBM	they have flexible work schedule	low salary bad health benefits no pension program	CEOs and management people please stop taking away our benefits and our money. Our salaries cannot afford gas price now.	Current Employee	Software Engineer	3	3	2	2	2	0	1	0	0	0	Negative	Negative	No opinion	TextTopic5_raw7	Long Work Hours / Workload Pressure
IBM	Global company, lots of career opportunities, lots of challenges business & technical, lots of chances to learn new things.	Very beaurocratic company, the procedures can be an obstacle in the way of doing one's job. Can be very frustrating due to the size of the organization which translates in a very slow reaction to changes. Can be diffcult to find the right people to help solve a	IBM is a good company to work for ... but be aware if the pittfalls of a huge organization.	Current Employee		4	3	4	4	3	0	1	0	0	0	Positive	Negative	No opinion	TextTopic5_raw2	Poor Management & Pay Issues
IBM	Great people to work with.	Could use more work/life balance.	Overall, a good place to work.	Current Employee	Consultant	3	3	4	3	4	0	1	0	0	0	Positive	Positive	No opinion	TextTopic5_raw8	Positive Work/Project Feedback
IBM	Great people, Easy pace of work, Wide range of jobs within IBM	Stagnation, slow deterioration of skills	Challenge to IBM's Leadership to motivate	Current Employee	Senior Software Engineer	5	3	2	3	1	0	1	0	0	0	Negative	Positive	No opinion	TextTopic5_raw1	Promotion & Salary Dissatisfaction
IBM	Even if individual managers do not nescessarily understand the rights of their employees (some do, some don't, like in any large org) systemically the rights of the employees seem protected. A far sight from perfect, but better than any other place Ive worked.	IBM is, as it has always been, a place that it takes a long time for the average guy to make good money. Ive seen many young managers handed great responsibility while making far less then their run of the mill employees. Also, the pension isn't any	IBM Still an OK Place to Work	Current Employee	Software Engineer	3	3	4	3	4	0	1	0	0	0	Positive	Mild	No opinion	TextTopic5_raw5	Salary Comparison & Pay Equity

firm	pros	cons	headline	current	job_title	work_life_balance	culture_values	career_opp	comp_benefits	senior_mgmt	work_life_balance_missing	culture_values_missing	career_opp_missing	comp_benefits_missing	senior_mgmt_missing	recommend_label	ceo_approv_label	outlook_label	Top_Topic_Label	Final_Top_Topic_Label
IBM	Management has been good at allowing a flexible work schedule.	Billable hours are their product. To increase profit they need to increase billable hours.	Not a bad company.	Current Employee	IT Specialist	3	3	3	4	2	0	1	0	0	0	Negative	No opinion	No opinion	TextTopic5_raw3	Work-Life Balance

Research Questions – Aligned with Final Model – Unsupervised Learning

- Sentiment Trends
  The topic modeling output revealed consistent themes like Work-Life Balance, Compensation Dissatisfaction, and Career Growth Challenges. Although temporal trends weren't modeled explicitly, these themes reflect core sentiment areas across employee feedback.

- Industry Comparison
  Multiple models (Oracle, IBM, JPMorgan, MCD) were evaluated, each representing company-specific topic structures. The IBM model, with the lowest misclassification rate, suggests stronger and more coherent topics, indicating it captures more relevant satisfaction factors in that domain.

- Key Themes in Reviews
  Using the Text Topic node, 8 dominant topics were extracted and labeled (e.g., Positive Project Feedback, Long Work Hours, Poor Management). These reflect the most frequent positive and negative aspects across diverse employee experiences.

- Impact of CEO Approval
  While CEO approval was not directly modeled, some topics like Frustration with Management and Career Growth Challenges indirectly reflect employee trust and alignment with leadership, supporting further exploration into such correlations.

- Predictive Modeling
  A Gradient Boosting model using only the unsupervised topic scores achieved ~61% accuracy, confirming that textual feedback alone can predict overall_rating reasonably well — answering this research question affirmatively.

Expected Outcomes – Delivered

- The final Gradient Boosting model (Log + IDF) successfully predicted overall employee satisfaction ratings with accuracy ranging from 50% to 61% across companies.
- Company-specific supervised models highlighted variations in employee sentiment drivers across industries (tech, finance, food service).
- Integration of textual features (via Text Filter and Text Cluster) with structured ratings significantly improved model performance.
- The results validate the effectiveness of NLP and machine learning in modeling employee feedback at scale.
- Model scoring on unseen reviews provided a practical demonstration of how HR teams can use such models for real-time sentiment analysis and workforce planning.

Conclusion:

This project successfully leveraged both unsupervised and supervised machine learning techniques within SAS Enterprise Miner to analyze employee reviews from Glassdoor. Through comprehensive text mining workflows, we uncovered valuable insights into employee sentiment, key workplace themes, and satisfaction drivers across four diverse companies, IBM, Oracle, J.P. Morgan, and McDonald's.

On the unsupervised learning side, techniques like Text Clustering and Topic Modeling allowed us to identify latent themes within review content, revealing distinct patterns in employee feedback. Segment profiling further helped interpret clusters by associating them with structured ratings such as work-life balance, compensation, and career opportunities.

For supervised learning, we applied predictive models—namely Decision Trees and Gradient Boosting—to classify overall employee satisfaction ratings. After experimenting with various text weighting schemes, the Gradient Boosting model using Log frequency and Inverse Document Frequency (IDF) emerged as the best performer, achieving the lowest misclassification rates across most companies. This validated our approach to feature engineering and model optimization.

Overall, the project not only demonstrated the power of text mining in human resource analytics but also highlighted how structured and unstructured data can be integrated effectively to deliver actionable insights for companies and job seekers alike.

References:

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